

ERIC SMITH

CITIZEN OF USA

CAREER OVERVIEW

Data mining, AI and Machine Learning expert. Generative AI Solution Architect with LLM using LangChain, CrewAI, AutoGen Agent Frameworks. M.S. in Advanced Analytics from NC State University. 15 years of hands-on professional experience as a Data Scientist & AI/ML Data Architect. Experienced in leading teams for AI Platform development in the Cloud. Expert in NLP/NLU, various Gen AI technologies including multi-modal embedding, Vector knowledge databases, RAG Agentic solutions and fine-tuning of LLMs. Experienced in AWS, Azure and GCP. Ph.D. in Computing Engineering with 10 US patents.

EDUCATION (2010)

- M.S. in Advanced Analytics from North Carolina State University, Raleigh, NC, USA

WORK EXPERIENCE (2010 - 2025)

- **2025/1 – present (Major Multinational Supermarket Chain): Lead AI Solution Architect and Principal Data Scientist** of a major retail business for fixing their data anomalies by introducing advanced machine learning techniques. Also leading the development of automation for IT process improvement of Retail Demand Forecasting (RDF) and Advanced Inventory Planning (AIP) systems.
- **2023/12 – 2024/12 (Volkswagen, USA): Gen AI Solution Architect and Lead Data Scientist** for a European Automobile company using Statistical as well as Neural Network based custom modeling, and OpenAI / GCP based LLMs using RAG Gen AI solutions. The project was on predicting vehicle conditions based on customer feedback and complaints received at their Call Centers.
- **2022/7 – 2023/11 (OPTUM / UHG): AI/ML Solution Architect** at OPTUM Health Care. Developed NLP (Natural Language Processing) based customer call center analytics using pretrained word embedding models.
- **2021/7-2022/6 (Minerva Analytics Inc.): ML Algorithm and AI Solution Architect** in Minerva Analytics Inc. Developed an image and vision analysis system for anomaly detection targeting video surveillance.
- **2020/7 – 2021/6 (Janssen / J&J): Senior AI/ML Consultant** at Janssen Pharmaceuticals (J&J) - built a predictive model-based optimization system for maximizing sales while satisfying various business constraints.
- **2018/1 – 2020/6 (Omniceil, Inc.): Technical Lead, AI/ML Solution Architect and Senior Data Scientist** at Omniceil for creating enterprise level data migration, analytics, machine learning and AI for the cloud (AWS / Azure), Proof of Concept (PoC), Data Architecture, Data Engineering, Data Security and Privacy, Model Deployment, Model life cycle and maintenance. Focused on system level scalability and performance.

- **2010 – 2017 (Bank of America Corporation): AI/ML Solution Architect & Data Scientist** at Bank of America and AT&T. Developed business intelligence for recommending actionable insights - thereby enhancing revenue / profit while managing risk.

CORE COMPETENCIES AS A DATA SCIENTIST

- Strong capability of analyzing a business problem and transforming it into a form that can be analyzed statistically and generate predictive models and business intelligence to provide actionable insights and strategic recommendations.
- Various statistical techniques including Predictive Modeling, Linear Regression, Logistic Regression, Decision Trees, XGBoost, GLM, Time Series Forecasting, Clustering, Hypothesis tests, Optimization.
- Big Data analysis and data mining for KPI dash-boarding and reporting by Spark/Scala/SQL/PySpark.
- Proficient in Analytic Data Warehouse design and implementation including ETL pipeline.
- Several Statistical languages and tools including Python, SAS, Rapid Miner.
- Advanced modeling techniques, including Neural Networks and Deep Learning for NLP, image and video analysis and modeling including GCP Gemini and Model Garden technologies.
- Proficient in using Transfer Learning based pre-trained Word Embedding technologies, e.g., Hugging Face LLM models including GPT, BERT, etc. for NLP based advanced text analytic modeling.
- Familiar with Enterprise architecture by TOGAF and AI/ML frameworks LangChain and LlamaIndex.

SKILLSET

- **Cloud technologies:** AWS, Azure, GCP, Spark/Scala/SQL/PySpark, DataBricks, VertexAI, SageMaker.
- **Machine Learning & Artificial Intelligence (AI) technologies:** Supervised & Unsupervised Learning, Deep Learning Neural Networks, image and video modeling by CNN, KERAS / TensorFlow, Natural Language Processing/Understanding/Generation (NLP/NLU/NLG) by RNN/LSTM (Recurrent Neural Network), Deployment by Docker, ECS, EKS, AKS, REST micro-services, MLOPS technologies.
- **Industry experience of introducing analytic systems:** Banking, Insurance, Telecom, Healthcare, Pharmaceuticals, Retail, Federal, Casinos, Pets and Livestock.
- **Software engineering and R&D:** Java, C/C++, Micro-Services, Feasibility study and POC (Proof of Concept) prototyping for analytic data warehousing of machine learning for cloud-based solutions, Data Engineering, UML, System & Platform Integration, Agile Development, GIS.

PATENTS

- Inventor/co-inventor of 10 US technology patents.